# The Hardy–Weinberg Law

50A

It should come as no surprise that the two men responsible for this law are named Hardy and Weinberg. In 1908, Godfrey Hardy (an English mathematician) and Wilhelm Weinberg (a German Doctor) independently described how, under certain conditions, predictions can be made about genotypes that will arise in a population.

The Hardy–Weinberg Law assumes that:

- the breeding population is sufficiently large
- other populations of the same species don't interbreed with the population studied
- mating is random
- all individuals are equally fertile and physically able to breed
- no mutation is occurring.

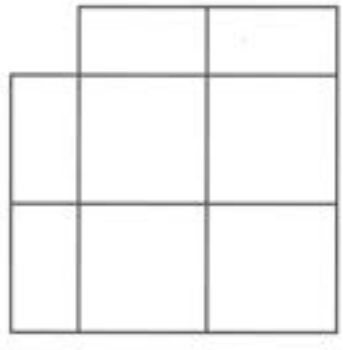
> The ratios of alleles in the gene pool of a population will not be changed by sexual recombination, but only by outside influences.

**1** Comment on one of the above assumptions of the Hardy–Weinberg Law.

_____

_____

A Punnett square can be used to calculate the inheritance of a single trait. It shows us how heterozygous crosses produce 75% of offspring carrying the dominant allele and 75% carrying the recessive allele. These figures might seem impossible, they add up to 150%, but are correct.



**2** In the space provided on the right, draw a Punnet square showing the cross Aa × Aa.

**3** Use your results to explain why the statement '*75% of offspring carry the dominant allele and 75% carry the recessive allele*' is true. _____

_____

Another tool, which can be used for more than one trait, is the equation:

$$p + q = 1$$

In this equation $p$ represents the dominant allele, $q$ represents the recessive allele and 1 stands for 100% of the population.

*Drosophila melanogaster* is a species of fruit fly. The species is infamous in Australia for causing damage to fruit crops but, for many years it has been an invaluable tool for scientists studying genetics. The main reason is that a *Drosophila melanogaster individual* can be born, live, breed and die within a period as short as two weeks. These fruit flies also have a considerable number of genes in common with humans.

**4** Explain why the short life cycle of *Drosophila melanogaster* makes it a favourite with geneticists.

_____

## Example

Imagine we are working with a population of 100 *Drosophila melanogaster* in a jar. This particular population contains the following genotypes: 30 AA, 60 Aa and 10 aa.

The proportion of dominant alleles (A) is calculated by substituting into the equation, $p + q = 1$, $p$ = 30 AA + (0.5 × 60) Aa = 60. Note: we use 0.5 Aa because only one allele in the pair is the dominant allele.

The proportion of recessive alleles (a) is calculated as follows: $q$ = 10 aa + (0.5 × 60) Aa = 40.

Now check. Does $p + q = 1$?

Substitute values: 60 + 40 = 100 (or 100% of the population studied)

We have accounted for the complete population, so $p + q = 1$.



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50B

## Time to do some sums!

Suppose that jar 'A' contains fruit flies with these genotypes: 29 AA, 44 Aa and 16 aa.

5 What is the total number of fruit flies in jar 'A'? _____

6 Using the formula given on work sheet 50A, calculate $p$. Show your working.

_____

_____

Often, before embarking on a calculation, we make an estimate to gain a rough idea of the answer.

7 Do some quick mathematics to estimate what you expect $q$ to be.

_____

8 Now calculate $q$, showing your working. _____

_____

Jar 'B' contains fruit flies with the following genotypes: 20 AA, 18 Aa and 7 aa.

9 What is the total number of fruit flies in jar 'B'? _____

10 Calculate $p$, showing your working. _____

_____

11 Calculate $q$, showing your working. _____

_____

## Minding your *p*s and *q*s!

If you use a Punnett square to cross $pq$ and $pq$, you will find that:

- $\frac{1}{4}$ of offspring are homozygous dominant ($pp$)
- $\frac{1}{2}$ of offspring are heterozygous ($pq$)
- $\frac{1}{4}$ of offspring are heterozygous recessive ($qq$).

Another way to express this mathematically is shown on the right.

$$p^2 + 2pq + q^2 = 1$$
where 1 = 100%

This is the same as saying $(AA)^2 + 2\ Aa + (aa)^2 = 1$

12 Find the number that represents $pp$ (or $p^2$) for jar 'A' above. _____

13 Find the number that represents $2pq$ for jar 'A' above. _____

14 Find the number that represents $qq$ (or $q^2$) for jar 'A' above. _____

15 Find the number that represents $pp$ (or $p^2$) for jar 'B' above. _____

16 Find the number that represents $2pq$ for jar 'B' above. _____

17 Find the number that represents $qq$ (or $q^2$) for jar 'B' above. _____

18 What is meant by 'homozygous dominant'? _____

_____

19 What is meant by 'heterozygous'? _____

_____

20 What is meant by 'homozygous recessive'? _____

_____

21 'Fruit flies (*Drosophila melanogaster*) have approximately 14 000 genes.' True or false? _____

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